

CAUSAL EXAGGERATION: UNCONFOUNDED BUT INFLATED CAUSAL ESTIMATES

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ABSTRACT

This paper shows that the feature that makes causal identification strategies credible, discarding variation to isolate exogenous sources, can also reduce the reliability of resulting estimates. By discarding variation, causal strategies reduce statistical power, leading statistically significant estimates to exaggerate true effects. Through theory and calibrated simulations, I formalize this confounding-exaggeration trade-off and show that the exaggeration bias can exceed the confounding bias the causal strategy does eliminate. I then introduce a ready-to-use diagnostic tool to identify the identifying variation in an applied study: the observations actually driving the causal estimate.

KEYWORDS: Identifying Variation, Causal Inference, Statistical Power, Applied Microeconomics, Exaggeration, Simulations, Package

PROJECT'S WEBSITE: https://vincentbagilet.github.io/causal_exaggeration

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I INTRODUCTION

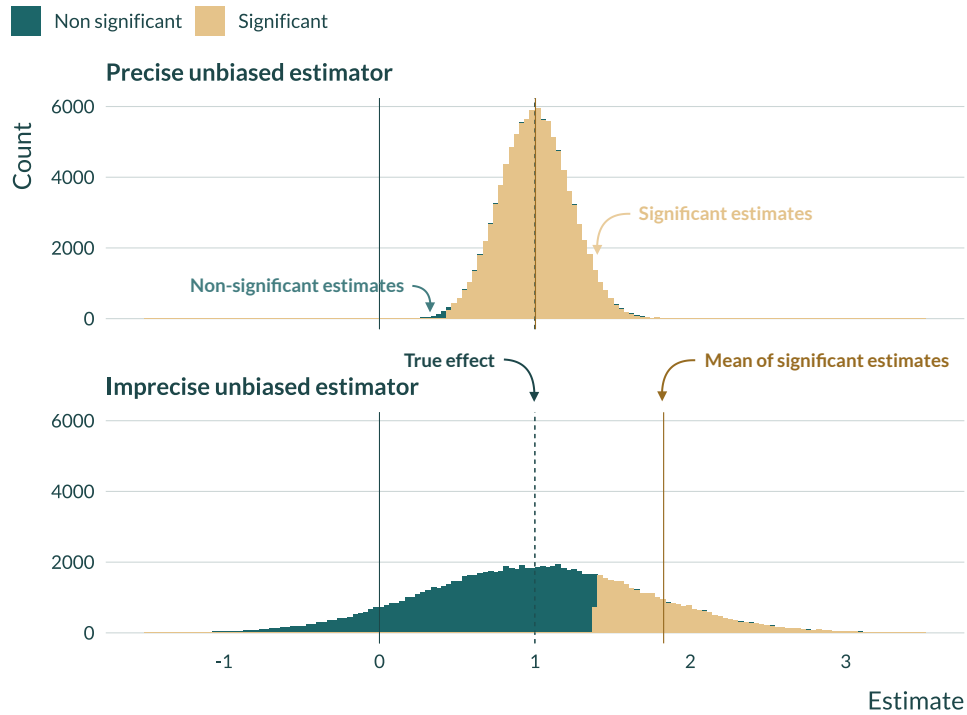
To isolate causal effects, identification strategies such as Regression Discontinuity (RD), Instrumental Variables (IV), Difference-in-Differences (DiD), and event-study designs only use a portion of the variation in the data. Though conceptually distinct, these strategies share a common structure: they all discard variation to focus on its exogenous part, whether by restricting the sample to observations near a threshold, retaining only the variation explained by an instrument, exploiting a set of exogenous shocks, or partialling out invariant components of the data. This discarding of variation makes these strategies credible, but it can also partly hide where identification comes from, so that identification may rely on much less variation than is apparent.

Building on this unifying perspective, this paper shows that when causal strategies discard a large share of the treatment variation, the identifying variation that remains may not be sufficient to precisely estimate the effect of interest, leading to exaggerated significant estimates. This generates a fundamental trade-off between reducing confounding and exaggerating the true effect size: improving identification can paradoxically pull published estimates away from the true effect. I formalize and document this confounding-exaggeration trade-off and introduce a ready-to-use diagnostic tool to identify the identifying variation in an applied study: the observations and sources of variation that actually drive the causal estimate.

To clarify the trade-off, I begin with an overview of the component that has received relatively less attention in economics: exaggeration. Exaggeration of significant estimates arises in imprecise studies because such estimates constitute a selected subsample of all estimates (Gelman and Tuerlinckx 2000). Figure 1 illustrates this phenomenon. Estimates that are statistically significant at the 5% level are at least 1.96 standard errors away from zero; when standard errors are large, they lie in the tails of the distribution of all possible estimates and thus exaggerate the true effect or can even be of the opposite sign. Statistical power therefore matters even after a statistically significant estimate has been obtained: when it is low, significant estimates from an unbiased estimator are necessarily far from the true effect, and the less precise the estimator, the larger the exaggeration. An estimator $\hat{\beta}$ of the true effect β might be unbiased in the traditional sense of $\mathbb{E}[\hat{\beta}] = \beta$ but conditionally biased in the sense that $\mathbb{E}[\hat{\beta} | \text{Significant}] \neq \beta$.

Exaggeration of published estimates occurs when two conditions arise simultaneously: 1) a publication bias favors a certain type of results, and 2) statistical power is low. Both features are observed in the economics literature. First, a large body of literature documents selection on significance in economics, whether through editorial decisions, specification choices, or the

Figure 1 – Distribution of Two Unbiased Estimators with Different Variances and Significance of Resulting Estimates



Notes: 100,000 draws from two normal distributions $\mathcal{N}(1, 0.05)$ and $\mathcal{N}(1, 0.5)$. Significance level: 5%

decision to pursue a project further (Rosenthal 1979, Brodeur et al. 2016, Andrews and Kasy 2019, Vivalt 2019, Abadie 2020, Brodeur et al. 2020). Importantly, the estimation biases resulting from selective publication and documented in Andrews and Kasy (2019) for instance only arise if power is low: when it is sufficiently large, most estimates are significant and favoring them does not lead to meaningful bias. Second, low statistical power is widespread in economics (Ioannidis et al. 2017, Ferraro and Shukla 2020, Young 2022). Ioannidis et al. (2017) finds that the median power of published studies across a wide range of areas in economics is only 18%, meaning that most studies, despite typically large sample sizes, are unlikely to detect the effects they target. As a result, they stress that nearly 80% of estimates are likely exaggerated by a factor of at least two. Consistent with this, DellaVigna and Linos (2022) shows that results from nudge experiments published in academic journals are about 6 times larger than those from large-scale experiments, a discrepancy the paper explains by low power, selection on significance, and characteristics of the experiments. It is therefore critical to take exaggeration into account and to understand its drivers, particularly when published estimates inform policy decisions. Yet while exaggeration is itself well documented, its drivers remain understudied.

The mechanism through which causal identification strategies can cause exaggeration

follows from the observation that, despite their apparent differences, these strategies can all be interpreted as forms of controlling, whether through their control function interpretation or by absorbing invariant dimensions of the data.¹ As such, it provides a unified way to characterize when the use of a causal strategy increases the variance of the estimator and thus the risk of exaggeration. Considering a simple linear homoskedastic model gives the intuition. Let $y_i = \alpha + \beta x_i + \delta w_i + u_i$, $\forall i \in \{1, \dots, n\}$, with x the variable of interest, w a potentially unobserved variable correlated with x and u , an error term. Under usual assumptions and using the Frisch-Waugh-Lovell (FWL) theorem, we get that $V_{\hat{\beta}_{omit}}$ and $V_{\hat{\beta}_{ctrl}}$, the variances of the estimators for β when omitting w and controlling for it are respectively:

$$\mathbb{V}_{\hat{\beta}_{omit}} = \frac{\sigma_{u_{omit}}^2}{n \sigma_x^2} = \frac{\sigma_{y^{\perp x}}^2}{n \sigma_x^2} \quad \text{and} \quad \mathbb{V}_{\hat{\beta}_{ctrl}} = \frac{\sigma_{u_{ctrl}}^2}{n \sigma_{x^{\perp w}}^2} = \frac{\sigma_{y^{\perp x, w}}^2}{n \sigma_{x^{\perp w}}^2}$$

where $\sigma_{u_{omit}}^2$ and $\sigma_{u_{ctrl}}^2$ are the variances of the residuals in the regression of y on x and of y on x and w respectively or equivalently the variances of the parts of y that are orthogonal to x and to x and w respectively ($\sigma_{y^{\perp x}}^2$ and $\sigma_{y^{\perp x, w}}^2$), σ_x^2 is the variance of x and $\sigma_{x^{\perp w}}^2$ is the variance of the part of x orthogonal to w . Thus,

$$\mathbb{V}_{\hat{\beta}_{omit}} < \mathbb{V}_{\hat{\beta}_{ctrl}} \Leftrightarrow \frac{\sigma_{y^{\perp x}}^2}{n \sigma_x^2} < \frac{\sigma_{y^{\perp x, w}}^2}{n \sigma_{x^{\perp w}}^2} \Leftrightarrow \frac{\sigma_{x^{\perp w}}^2}{\sigma_x^2} < \frac{\sigma_{y^{\perp x, w}}^2}{\sigma_{y^{\perp x}}^2}$$

Controlling for w increases the variance of the estimator if the fraction of the variance unexplained by w is greater for $y^{\perp x}$ than for x . Put differently, if controlling absorbs more of the variation in x than in the residual part of y ($y^{\perp x}$), the variance of the estimator increases. Since exaggeration increases with the estimator's variance, controlling for a confounder can increase exaggeration. Controls, including fixed effects, limit the identifying variation and, as a result, observations whose treatment status is largely explained by these controls contribute little, if at all, to identification. Identification may thus effectively rely on a small sample.

In the remainder of the paper, I first formally prove the existence of the trade-off for prevailing causal identification strategies. Specifically, I show that the bias caused by exaggeration can be larger than the one caused by confounders. I also analyze the drivers of exaggeration and demonstrate that it increases as the strength of the instruments decreases, the number of exogenous shocks decreases, or when controlling for a confounder absorbs more of the variation in the treatment than in the outcome. In this sense, the well-known bias-

1. Fixed Effects (FEs) based identification strategies such as DiD control for the invariant, unobserved, and arguably endogenous part of the variation in the outcome. An IV approach essentially partials out the variation in x unexplained by the instruments. Fuzzy-RD and propensity score matching can be thought of as control function approaches, of the forcing variable and propensity score respectively. In addition, excluding observations that are outside the bandwidth or unmatched is equivalent to controlling for observation-level fixed effects for these units.

variance trade-off is in fact a bias-bias trade-off: while confounding produces an unconditional bias on the estimator, the larger variance resulting from the use of causal identification can produce a conditional bias that exceeds this confounding bias.²

I then further investigate causal exaggeration in concrete settings where analytical results are less tractable. Monte Carlo simulations allow me to consider a broader range of identification strategies, more complex and realistic data generating processes, and to vary parameters of interest *ceteris paribus*. To make these simulations concrete, I calibrate them to emulate existing studies from education, environmental, health, labor, and political economics, deriving distributional parameters, sample sizes, and effect sizes from published work. A clear pattern emerges: as the identification strategy absorbs a larger share of the treatment variation, exaggeration increases sharply. These reductions in identifying variation can be substantial. In studies exploiting temporal variation in temperature, for instance, month fixed effects can absorb over 90% of the variation in high-temperature indicators. In designs exploiting discrete events, treated observations can represent less than 1% of the sample. For parameter values consistent with standard practice in the corresponding literatures, statistically significant estimates exaggerate the true effect by a factor of 1.3 to 4, despite unbiased estimators, correctly specified models, and valid identification.

Finally, I discuss how we, as applied researchers, can navigate the trade-off. Sensitivity analyses and power calculations can help evaluate the potential magnitude of confounding and exaggeration respectively. Considering the attention given to bias avoidance in the economics literature, making power central to non-experimental analyses, even after a significant estimate has been obtained, would help guard against exaggeration. Focusing more specifically on the drivers of the trade-off, I introduce an R package, `ididvar`, that facilitates the identification of the variation underlying causal estimates. Since causal identification strategies can generally be interpreted as forms of controlling, the treatment variation that remains after partialling out all controls, both standard covariates and those implied by the identification strategy, provides a unifying measure of the risk of exaggeration. The package compute each observation's contribution to the estimated treatment effect and characterizes the effective sample used for identification, without assumptions about the true effect size. In the illustrative application presented below, 70% of the nominal sample can be dropped without affecting the point estimate or standard error by more than 5%, suggesting that identification effectively relies on a much smaller sample than the reported one.

2. Throughout this paper, I use “bias” to refer to the expected deviation of estimates from the true effect, whether this deviation is unconditional and caused by confounding or conditional on significance and resulting from selective publication and imprecision. When using the term “bias” in “publication bias” however, I refer to yet another concept: the selective publication of certain types of results, which is distinct from any bias in the estimates themselves.

This paper contributes to four strands of the applied economics literature. First, the idea that causal identification estimators, while unbiased, may be imprecise is well understood as part of the bias-variance trade-off (Imbens and Kalyanaraman 2012, Deaton and Cartwright 2018, Hernán and Robins 2020, Ravallion 2020). I approach this trade-off through the prism of statistical power and publication bias. Limited precision resulting from the use of causal identification strategies not only makes it difficult to draw clear conclusions about the exact magnitude of effects, but, I argue, can also inherently lead to inflated published estimates, creating another form of bias.

Second, this paper contributes to the literature on statistical power and exaggeration. Studies in this literature rarely investigate the determinants of exaggeration or consider specific causal identification strategies separately (Ioannidis et al. 2017, Schell et al. 2018, Ferraro and Shukla 2020, Black et al. 2022, Stommes et al. 2023, Arel-Bundock et al. 2024). Existing frameworks for modeling and correcting publication bias, such as Andrews and Kasy (2019), typically focus on selection on significance, one of the two ingredients leading to estimation biases, and operate at the level of a body of literature by leveraging a distribution of published estimates. I focus on the other ingredient, low statistical power, which is by contrast directly shaped by design and modeling choices in individual studies. To my knowledge, only a limited number of studies explore broad determinants of exaggeration: Vu (2025) analyzes the impact of standard error corrections and in a companion paper, I study concrete design parameters that can generate exaggeration across a wide range of settings (Bagilet 2023). In the present paper, I link the exaggeration literature to core causal inference principles and propose an overarching mechanism: causal identification strategies discard variation, thereby increasing the risk of exaggeration. Further, the unifying interpretation of causal strategies as forms of controlling provides a common framework for understanding this mechanism across strategies and underlies the diagnostic tool introduced in this paper.

As such, this paper makes a methodological and practical contribution by facilitating the computation of concrete measures of each observation’s contribution to identification and, by extension, of the effective sample underlying the estimated effect. While the conceptual foundations for such measures exist (Angrist and Pischke 2009, Aronow and Samii 2016), the `ididvar` package specifically links them to causal identification strategies, implements and expands them. It allows applied researchers to compute, visualize, and compare these measures across specifications in a straightforward and consistent way, and to diagnose potential exaggeration risks *ex-post* without assumptions about true effect sizes. The package also provides a framework for identifying and communicating the set of observations contributing to identification and how this set changes across specifications, levels of fixed effects and sets of control variables.

Finally, this study contributes to the literature on replicability in economics (Camerer et al. 2016, Ioannidis et al. 2017, Christensen and Miguel 2018, Kasy 2021). The trade-off presented in the present paper suggests that convincing causal identification, while necessary, does not by itself shield estimates from replication threats. The tools discussed and introduced can be readily used in replication analyses or requested by referees to transparently document statistical power and the sources of identifying variation.

In the following section, I study the drivers of exaggeration and formally show in a simple setting that the use of causal identification strategies can exacerbate it. In Section III, I implement realistic Monte-Carlo simulations to illustrate the existence of the confounding-exaggeration trade-off. I discuss potential solutions to navigate this trade-off and introduce the `ididvar` package in Section IV and conclude in Section V.

II MATHEMATICAL DERIVATION

In this section, I characterize how causal identification strategies generate exaggeration through the variation they restrict for identification. I first recall the definition of the exaggeration ratio and show that it increases with the variance of the estimator. I then derive the asymptotic distributions of a series of estimators, from a long and a short OLS regression, an IV regression, and its reduced form, and show that the parameters governing each strategy directly determine the variance of the resulting estimator and thus the extent of exaggeration. This allows me to describe the mechanism through which each strategy generates exaggeration, and to show that, for any magnitude of OVB, there exist parameter values such that exaggeration from a causal identification strategy exceeds the confounding bias the strategy was designed to eliminate.

II. 1 PROPERTIES OF THE EXAGGERATION RATIO

Following Gelman and Carlin (2014), one can define the exaggeration ratio E , as the expectation of the absolute value of significant estimates over the absolute value of the true effect. For an estimator $\hat{\beta}$ of a true effect β , with standard error σ and a two-sided hypothesis test of size α with threshold value z_α , let

$$E(\hat{\beta}, \sigma, \beta, z_\alpha) = \frac{\mathbb{E} \left[|\hat{\beta}| \mid \beta, \sigma, |\hat{\beta}| > z_\alpha \sigma \right]}{|\beta|} \quad (1)$$

Lu et al. (2019) and Zwet and Cator (2021) showed that, for a given test and true effect sizes, the exaggeration ratio increases with the variance of an unbiased normally distributed

estimator. This proof can easily be extended to biased estimators, in particular with additively separable bias.³

Lemma 1. *For an estimator $\hat{\beta}_b \sim \mathcal{N}(\beta + b, \sigma^2)$ of a true effect of magnitude β and a fixed additively separable bias b of the same sign as the true effect,⁴*

- *E is a decreasing function of the Signal-to-Noise Ratio (SNR) $\frac{\beta}{\sigma}$, i.e. the relative precision of the estimator, and depends on σ only through this SNR.*
- $\lim_{\sigma \rightarrow \infty} E(\hat{\beta}_b, \sigma, \beta, z_\alpha) = +\infty$.

In other words, the larger the variance of the estimator, the more exaggeration and, in the limit, exaggeration is infinite. Figure 1 provides a clear intuition for this result in the unbiased case. The goal of this section is twofold: 1) to characterize the mechanism through which each identification strategy generates exaggeration, and 2) to show that this exaggeration can exceed any OVB in the corresponding naive regression. To study the former, given Lemma 1, one only needs to describe the evolution of the variances of these estimators with the parameters of interest: the IV strength in an IV setting, the number of exogenous shocks in a reduced form and the correlation between the explanatory variable of interest and the omitted variable of interest. Regarding the second objective, one can show that, for any given level of OVB, there exist parameter values for which the exaggeration from a causal identification strategy exceeds that OVB. These comparative statics isolate the role of design parameters in driving exaggeration, independently of any effect on confounding. In applied settings, parameter values are implicitly determined by the setting and the choice of instruments or control sets, which explains why they can differ across designs.

II. 2 SETTING AND DATA GENERATING PROCESS

Consider a usual linear homoskedastic regression model with an omitted variable. For any individual $i \in \{1, \dots, n\}$,

$$y_i = \beta_0 + \beta_1 x_i + \delta w_i + u_i \quad (2)$$

where y is the outcome, x the explanatory variable, w an unobserved omitted variable, u an unobserved error term. $(\beta_0, \beta_1, \delta) \in \mathbb{R}^3$ are unknown parameters. β_1 is the parameter of interest.

I assume homogeneous treatment effects and homoskedasticity, along with the usual OLS assumptions (*i.i.d.* observations, finite second moments, positive-definiteness of $\mathbb{E}[x_i x_i']$)—

3. All the proofs of the lemma and theorems are reported in appendix A.

4. I focus on cases where bias and true effect have the same sign as this represents the conservative scenario where exaggeration and OVB do not offset each other.

with $\mathbf{x}_i = (1, x_i)'$ — and u_i conditional mean-zero and uncorrelated with x_i and w_i). Assume that w_i is unobserved, correlated with x_i and that $\delta \neq 0$. To simplify the derivations, I further assume that the unobserved variable is centered, *i.e.* $\mathbb{E}[w_i] = 0$. I also assume that the variance of the component of w_i that is orthogonal to x_i (denoted $w_i^{\perp x}$) does not vary with x_i , *i.e.*, $\text{Var}(w_i^{\perp x} | x_i) = \text{Var}(w_i^{\perp x})$. Consider the following data generating process for x_i :

$$x_i = \mu_x + \gamma w_i + \epsilon_i \quad (3)$$

where $\gamma \in \mathbb{R}^*$ since x and w are correlated. Set $\rho_{xw} = \text{corr}(x, w) = \frac{\gamma\sigma_w}{\sigma_x}$. In the IV and reduced form sections, I further assume that there exists a valid instrumental variable z_i for x_i , *i.e.* that $\mu_x + \epsilon_i = \pi_0 + \pi_1 z_i + e_i$ where $(\pi_0, \pi_1) \in \mathbb{R}^2$ are unknown parameters. The existence or not of this valid instrument does not affect the results in the controlled and OVB cases. Since the instrument is valid, it satisfies exogeneity, *i.e.* $\mathbb{E}[z_i u_i] = 0$ and $\mathbb{E}[z_i w_i] = 0$, relevance, *i.e.* $\text{rank}(\mathbb{E}[z_i z_i']) = 2$, and positive-definiteness of $\mathbb{E}[z_i z_i']$. The data generating process for x_i becomes:

$$x_i = \pi_0 + \pi_1 z_i + \gamma w_i + e_i \quad (4)$$

I assume that e_i is uncorrelated with z_i and w_i , *i.e.* $\mathbb{E}[z_i e_i] = 0$ and $\mathbb{E}[w_i e_i] = 0$. I also assume homoskedasticity for this term, such that $\mathbb{E}[e_i^2 | z_i, w_i] = \sigma_e^2$ is constant.

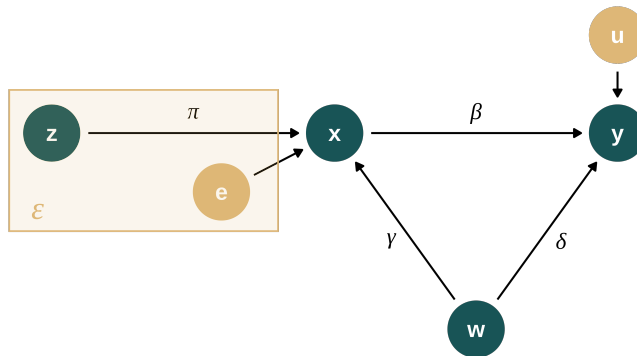
Overall, this DGP is close to the usual textbook one but with an additional omitted variable. For ease of exposition, I focus on a simple setting with a limited number of variables, though the results directly extend to higher-dimensional settings. As such, the present setting represents a simple baseline. Complexifying it, relaxing assumptions or accounting for aspects such as clustering or heteroskedasticity would typically increase estimator variance and thus exaggeration. The results presented here can therefore be viewed as lower bounds of the confounding-exaggeration trade-off. The Directed Acyclic Graph (DAG) in figure 2 represents the data generating process.

II. 3 ASYMPTOTIC DISTRIBUTIONS OF THE ESTIMATORS

For each model, the goal is to study the evolution of the sampling distribution variances with the value of the parameters of interest, for a fixed OVB. I assume that sample sizes are large enough that the sampling distributions are well approximated by the asymptotic distributions.⁵ To make comparisons across settings meaningful, I hold the variances of the

5. In the IV case, convergence requires instruments not to be arbitrarily weak relative to the sample size. When instruments are very weak, finite-sample distributions may differ substantially from asymptotic

Figure 2 – DAG of the Data Generating Process



Notes: for clarity the error terms are represented in this graph, in beige. Model parameters are noted as edge labels.

“defined” variables ($\sigma_y^2, \sigma_x^2, \sigma_w^2$, and σ_z^2) fixed. When varying the parameters (γ, δ , or π_1), I adjust the error variances (σ_u^2 and σ_e^2) so that the variances of the defined variables do not vary with the parameters. Only the proportions of their variance unexplained by the models do.

II. 3.1 NAIVE REGRESSION (OVB)

First, let us study the benchmark against which to compare the causal approaches and show that, for a given bias, its asymptotic variance does not depend on the parameters that will vary in other specifications (γ or π_1). Consider the “naive” regression of y on x (with w omitted).

Lemma 2. *Based on the data generating process described in Section II. 2, for $\hat{\beta}_{\text{OVB}}$ the OLS estimate of β_1 in the regression of y on x , $\sqrt{n}(\hat{\beta}_{\text{OVB}} - \beta_1) \xrightarrow{d} \mathcal{N}(b_{\text{OVB}}, \sigma_{\text{OVB}}^2)$, with*

$$b_{\text{OVB}} = \frac{\delta\gamma\sigma_w^2}{\sigma_x^2} \quad \text{and} \quad \sigma_{\text{OVB}}^2 = \frac{\sigma_u^2 + \delta^2\sigma_w^2(1 - \rho_{xw}^2)}{\sigma_x^2}$$

The intuition for the formula of the asymptotic variance has been discussed in the introduction: $\sigma_u^2 + \delta^2\sigma_w^2(1 - \rho_{xw}^2)$ is the part of the variance in y that is not explained by x ($\sigma_{y \perp x}^2$).

Since σ_x^2 and σ_w^2 are fixed, reasoning at $b_{\text{OVB}} = cst$ is equivalent to considering that $\gamma\delta = \kappa = cst$. Noting that $\sigma_y^2 = \beta_1^2\sigma_x^2 + \delta^2\sigma_w^2 + 2\beta_1\kappa\sigma_w^2 + \sigma_u^2$, we can rewrite the variance of the estimator as

$$\sigma_{\text{OVB}}^2 = \frac{\sigma_y^2 - \beta_1^2\sigma_x^2 - 2\beta_1\kappa\sigma_w^2 - \kappa^2\frac{\sigma_w^4}{\sigma_x^2}}{\sigma_x^2}$$

distributions, typically exhibiting fatter tails. However, when power is low, significant estimates are located in the tails of the distribution and the additional mass in the tails leads to larger exaggeration.

For a fixed bias ($\kappa = cst$), σ_{OVB}^2 does not vary with γ and, based on Lemma 1, nor does E_{OVB} .

II. 3.2 CONTROLLED REGRESSION

Next, let us turn to the “ideal” case in which no variable is omitted, *i.e.* we control for the omitted variable w and thus partial out confounders. This model accurately represents the DGP. This corresponds to the usual OLS setting with a constant and two regressors that are uncorrelated with the error: y regressed on x and w .

Lemma 3. *Based on the data generating process mentioned previously, for $\hat{\beta}_{\text{CTRL}}$ the OLS estimator of β_1 in the regression of y on x and w , $\sqrt{n}(\hat{\beta}_{\text{CTRL}} - \beta_1) \xrightarrow{d} \mathcal{N}(0, \sigma_{\text{CTRL}}^2)$, with*

$$\sigma_{\text{CTRL}}^2 = \frac{\sigma_u^2}{\sigma_x^2(1 - \rho_{xw}^2)}$$

$\sigma_x^2(1 - \rho_{xw}^2)$ is the part of the variance of x that is not explained by w ($\sigma_{x \perp w}^2$) and σ_u^2 the part of the variance of y that is not explained by x nor w ($\sigma_{y \perp x, w}^2$); here too we retrieved a result described in the introduction. For a given bias, we can rewrite σ_{CTRL}^2 as a function of fixed variances and one varying parameter, γ :

$$\sigma_{\text{CTRL}}^2 = \frac{\sigma_y^2 - \beta_1^2 \sigma_x^2 - \frac{\kappa^2}{\gamma^2} \sigma_w^2 - 2\beta_1 \kappa \sigma_w^2}{\sigma_x^2 - \gamma^2 \sigma_w^2}$$

Since the numerator and denominator respectively increase and decrease with $|\gamma|$, σ_{CTRL}^2 increases with $|\gamma|$. For a given bias, the more w is correlated with x (and thus roughly the less it is with y since $\delta\gamma = const$), the larger the variance of the estimator. In addition, we can note that, for a given bias, the variance of the estimator can be arbitrarily large since $\lim_{\gamma^2 \rightarrow \frac{\sigma_x^2}{\sigma_w^2}} \sigma_{\text{CTRL}}^2 = +\infty$.

II. 3.3 INSTRUMENTAL VARIABLES

In the previous section, I considered a case in which we removed variation that included unwanted endogenous variation. Let us now turn to the IV, a converse situation where we select variation to keep, exogenous variation. Consider the IV model in which y is regressed on $\mathbf{x}_i = (1, x_i)'$ instrumented by $\mathbf{z}_i = (1, z_i)'$. This is a just-identified case and $\hat{\beta}_{\text{2SLS}} = \hat{\beta}_{\text{IV}}$.

Lemma 4. *Based on the data generating process mentioned above, for $\hat{\beta}_{\text{IV}}$ the IV estimator*

of β_1 in the regression of y on x instrumented by z , $\sqrt{n}(\hat{\beta}_{IV} - \beta_1) \xrightarrow{d} \mathcal{N}(0, \sigma_{IV}^2)^6$, with

$$\sigma_{IV}^2 = \frac{\sigma_u^2 + \delta^2 \sigma_w^2}{\sigma_x^2 \rho_{xz}^2}$$

The numerator is $\sigma_{y \perp \hat{x}}^2$, the part of the variance in y that is not explained by $\hat{x}$, the predicted value of x in the first stage and the denominator is $\sigma_{\hat{x}}^2$. For a given bias, noting that $\rho_{xz} = \text{corr}(x, z) = \pi_1 \frac{\sigma_z}{\sigma_x}$ and replacing σ_u^2 , we can rewrite σ_{IV}^2 as a function of fixed variances and one varying parameter, π_1 :

$$\sigma_{IV}^2 = \frac{\sigma_y^2 - \beta_1^2 \sigma_x^2 - 2\beta_1 \kappa \sigma_w^2}{\pi_1^2 \sigma_z^2}$$

The smaller π_1 , the larger σ_{IV}^2 . In addition, $\lim_{\pi_1 \rightarrow 0} \sigma_{IV}^2 = +\infty$.

II. 3.4 REDUCED FORM

Let us now consider the regression of the outcome of interest directly on the instrument. Plugging equation 4 into equation 2 yields:

$$y_i = (\beta_0 + \beta_1 \pi_0) + (\beta_1 \pi_1) z_i + ((\delta + \beta_1 \gamma) w_i + u_i + \beta_1 e_i)$$

Note that the resulting effect size is different from that of the other models. To make them comparable, one can set π_1 to 1 so that an increase of 1 in the instrument causes an increase of β_1 in y . Regardless of this assumption, regressing y on z corresponds to the usual bivariate, unbiased case and directly gives the following result:

Lemma 5. *Based on the data generating process mentioned previously, for $\hat{\beta}_{RED}$, the OLS estimator of the reduced form regression of y on z , $\sqrt{n}(\hat{\beta}_{RED} - \beta_1) \xrightarrow{d} \mathcal{N}(0, \sigma_{RED}^2)$, with*

$$\sigma_{RED}^2 = \frac{\sigma_y^2 - \beta_1^2 \pi_1^2 \sigma_z^2}{\sigma_z^2}$$

The numerator is the part of the variance of y that is not explained by z ($\sigma_{y \perp z}^2$). In addition, the smaller σ_z^2 , the larger σ_{RED}^2 and $\lim_{\sigma_z \rightarrow 0} \sigma_{RED}^2 = +\infty$.

In the binary case, $\sigma_z^2 = p_1(1 - p_1)$ with p_1 the proportion of treated observations, *i.e.*, the proportion of 1 in z . When most observations have the same treatment status, *i.e.*, p_1

6. When instruments are arbitrarily weak however, the asymptotic distribution is non-normal, with fat tails. This type of distribution increases rather than decreases exaggeration, as underlined above.

close to 0 or 1, σ_z^2 tends to zero and σ_{RED}^2 diverges. There is not enough variation in the treatment status to precisely identify the effect of interest.

II. 4 EXAGGERATION RATIOS

Combining the results from Lemma 2 through 5 regarding the asymptotic distribution of the various estimators with Lemma 1 stating that exaggeration increases with the variance of a normally distributed estimator yields:

Theorem 1. *For the data generating process described in Section II. 2, the exaggeration ratio of the controlled, IV and reduced form estimators, respectively E_{CTRL} , E_{IV} and E_{RED} , are such that:*

- E_{CTRL} increases with the correlation between the omitted variable and the explanatory variable of interest (i.e. $|\gamma|$ or $|\rho_{xw}|$), for a given bias,
- E_{IV} decreases with the strength of the IV (i.e. with $|\pi_1|$ or $|\rho_{xz}|$),
- E_{RED} increases, in the binary case, when the number of exogenous shocks decreases

Each of these parameters determines the subset of variation in x leveraged for identification: the variation orthogonal to controls, that is explained by the instruments or that results from shocks. Each identification strategy restricts this variation, and the magnitude of restriction determines exaggeration. In other words, the same feature that makes each strategy credible, restricting variation to its exogenous component, is also what generates the risk of exaggeration. Using the same lemma and the limit properties of the variances described in Section II. 2, and since, at fixed bias, E_{OVB} does not vary with the parameters of interest, we get:

Theorem 2. *For the data generating process described in Section II. 2, $\forall b_{\text{OVB}}$,*

- $\exists \gamma$ s.t. $E_{\text{CTRL}} > E_{\text{OVB}}$
- $\exists \pi_1$ s.t. $E_{\text{IV}} > E_{\text{OVB}}$
- $\exists \sigma_z$ s.t. $E_{\text{RED}} > E_{\text{OVB}}$

In any setting, i.e. for any OVB, some causal designs can yield statistically significant estimates which are more biased than that of a naive regression. In other words and through the prism of identifying variation: causal identification strategies can remove so much variation that exaggeration becomes larger than the bias resulting from omitting any variable.

III SIMULATIONS

To illustrate how each identification strategy generates exaggeration and to expand the results of the formal mathematical analysis to more complex settings and identification strate-

gies, I build simulations calibrated to real-world examples from economics of education for RDD, political economy for IV, health economics for exogenous shocks, and environmental and labor economics for control and fixed effects approaches. I split the simulations by identification strategy, as each strategy restricts variation through a distinctive channel. Rather than reproducing a specific study, I calibrate my simulations to emulate a typical non-experimental study from each literature and compare their Signal-to-Noise Ratios (SNR) to the estimate/standard error ratios in the corresponding literatures. As underlined by Lemma 1, the SNR is a sufficient statistic for the exaggeration ratio; for the exaggeration of the simulation to be representative of actual studies, we only need to calibrate its SNR. I make conservative modeling choices throughout, considering simple linear models with constant and homogenous treatment effects, *i.i.d.* observations, homoskedastic errors and correctly specified models so that the results are likely a lower bound for exaggeration. Extensively documented code is available on the [project’s website](#).

III. 1 REGRESSION DISCONTINUITY DESIGN

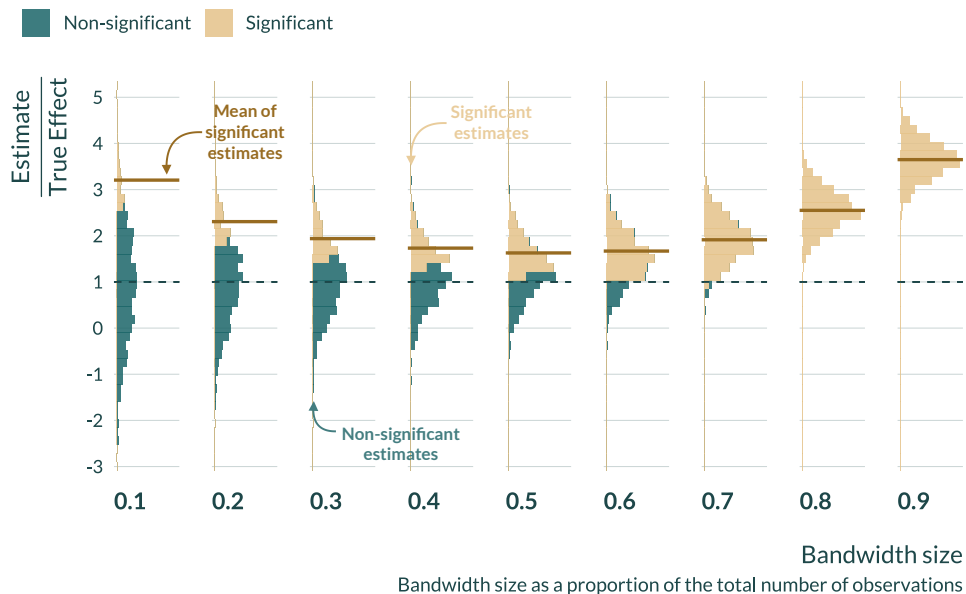
Intuition. The RDD achieves identification by restricting estimation to observations close to the threshold, discarding those further away whose comparability cannot be assumed. This restriction is what makes the design credible, but it also reduces the effective sample available for identification. A smaller bandwidth means a smaller effective sample and lower precision, increasing the risk of exaggeration. The confounding-exaggeration trade-off is thus mediated by the bandwidth size.

Case-study and simulation procedure. I consider a standard application of the sharp RD design from economics of education in which students are assigned to additional lessons based on the score they obtained on a standardized test.⁷ Students whose initial score falls below a threshold receive the treatment. Since students far from the threshold may differ in unobserved ability, the RDD compares outcomes only for students close to the cutoff. In the simulations, treatment assignment is sharp: $T_i = \mathbb{I}[Qual_i < C]$, where $Qual_i$ is the initial test score and C the cutoff. Unobserved ability w_i enters both scores non-linearly via a cubic function f : $Qual_i = \mu_q + \gamma f(w_i) + \epsilon_i$ and $Final_i = \beta_0 + \beta_1 T_i + \eta Qual_i + \delta f(w_i) + u_i$. Confounding from w_i shrinks as the bandwidth around C narrows, since students within a narrow window are similar in ability. I derive the distribution of grades from Department of Education statistics, effect sizes from a meta-analysis of RCTs ([Kraft 2020](#)), and sample and bandwidth sizes from an existing RDD study on a similar question ([Jacob and Lefgren](#)

7. [Thistlethwaite and Campbell \(1960\)](#) introduced the RDD in a similar setting.

2004). Further calibration details are on the [project's website](#). I generate 1000 datasets of 60,000 observations and estimate the treatment effect for different bandwidth sizes.

Figure 3 – Evolution of the Bias with Bandwidth Size in the Regression Discontinuity Design Simulations, Conditional on Significance



Notes: 1000 iterations. Significance level: 5%, $N = 60,000$. The brown lines represent the average of significant estimates. The bandwidth size is expressed as the proportion of the total number of observations in the entire sample. Details on the simulation are available at this [link](#).

Results. Figure 3 displays the results of these simulations. For large bandwidths, OVB biases the estimates. As the bandwidth shrinks, the OVB disappears but the effective sample size decreases, widening the distribution and pushing statistically significant estimates into the tails. The average of all estimates converges to the true effect, but the average of significant estimates never does. This is analogous to the optimal bandwidth literature's bias-precision trade-off (Imbens and Kalyanaraman 2012), recast here as a confounding-exaggeration trade-off. For a bandwidth size consistent with those in the literature on which the simulations are calibrated, significant estimates would be 3.2 times larger than the true effect.⁸

III. 2 FIXED EFFECTS AND CONTROLLING FOR CONFOUNDERS

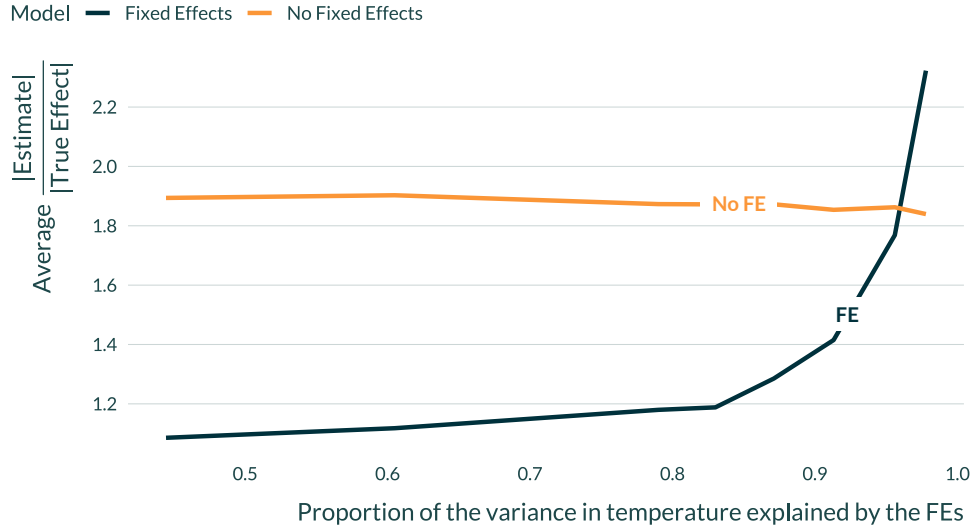
Intuition. Fixed effects and controls achieve identification by absorbing variation shared between the treatment and omitted variables. This is precisely what makes them credible

⁸ A typical article studying a similar question has an optimal bandwidth yielding roughly 5000 observations, which corresponds to a bandwidth of 0.1 in my simulations.

for causal identification. Yet, as shown in the introduction and Section II, if they absorb more of the variation in the treatment than in the outcome, they increase the variance of the estimator and can lead to exaggeration. The confounding-exaggeration trade-off is mediated by the share of treatment variation absorbed by the controls and fixed effects.

Case-study and simulation procedure. I consider studies on the impact of temperature on worker productivity, a literature that typically relies on high-dimensional fixed effects models (Lai et al. 2023, for a review). Formally, temperature at time t in period τ follows $Temp_{t\tau} = \mu_{Temp} + \gamma\lambda_\tau + \epsilon_t$, where λ_τ are period fixed effects with mean 0 and variance 1, so that γ governs the share of treatment variance they explain. Worker productivity is $Prod_{it\tau} = \beta_0 + \eta_i + \delta\lambda_\tau + \beta_1 \cdot \mathbb{1}\{Temp_{t\tau} \in (T_L, T_H]\} + u_{it}$, where η_i are individual fixed effects and $(T_L, T_H]$ the temperature bin of interest. The period shock λ_τ enters both temperature and productivity, generating confounding when period fixed effects are omitted; I hold $\kappa = \gamma\delta$ constant as γ varies so that the OVB remains unchanged. I calibrate the simulations to a typical study from this literature (Stevens 2017, Somanathan et al. 2021, Lai et al. 2023, LoPalo 2023, among others); further details are on the [project's website](#). For each value of γ , I generate 1000 datasets of 49,000 observations and estimate two models, one with and one without period fixed effects.

Figure 4 – Evolution of the Bias of Statistically Significant Estimates with the Proportion of the Variance in Temperature Explained by the Fixed Effects, in the FE Simulations



Notes: 1000 iterations. Significance level: 5%, $N = 49,000$. The blue line indicates the average bias ratio for statistically significant estimates from the model including period FEs. The orange line represents the same quantity but for the model without the fixed effects. Details on the simulation and calibration are available at this [link](#).

Results. Figure 4 displays the results of these simulations. When the (period) fixed effects explain a large part of the variation in temperature, the estimator is imprecise and leads to exaggeration to the point that it becomes larger than the simulated OVB. In actual settings, month fixed effects explain up to 90% of the variation in temperature, a value associated with an exaggeration factor of 1.3 in these simulations. However, the simulations do not model location fixed effects, which would likely further increase exaggeration. In addition, this setting features relatively large sample sizes; settings with a smaller one would likely exhibit substantial exaggeration even when fixed effects absorb a smaller share of variation.

III. 3 INSTRUMENTAL VARIABLES STRATEGY

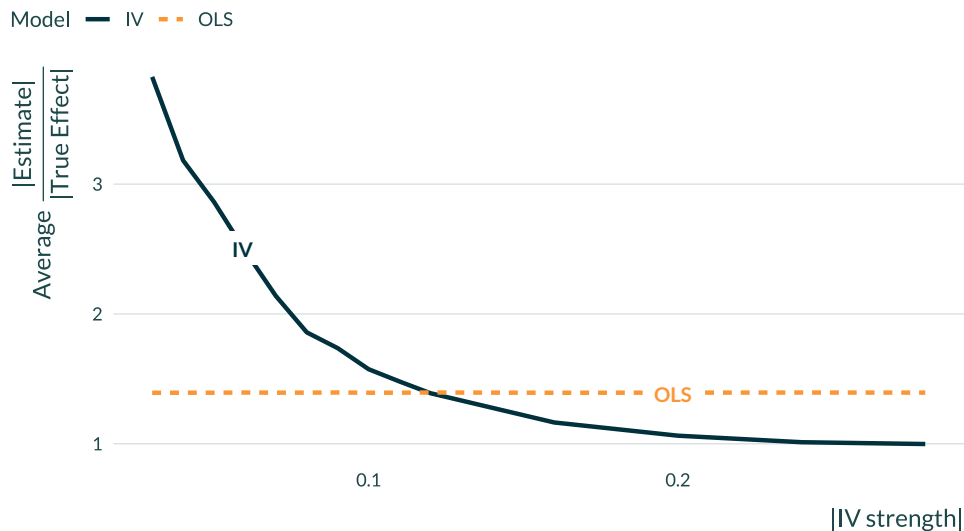
Intuition. The IV achieves identification by isolating the exogenous fraction of treatment variation, *i.e.* the variation explained by the instrument. This selection is what allows the IV to eliminate confounding. Yet the same restriction generates exaggeration: even when the exogenous fraction is sufficient to identify the causal effect on average, a weak instrument leaves the estimator imprecise and statistically significant estimates substantially inflated. The confounding-exaggeration trade-off is thus mediated by instrument strength.

Case-study and simulation procedure. I consider the impact of voter turnout on election results, instrumented by rainfall on election day.⁹ The DGP is: $Share_i = \beta_0 + \beta_1 Turnout_i + \delta w_i + u_i$, where $Share_i$ is the Republican vote share, w_i an unobserved confounder, and $Turnout_i = \pi_0 + \pi_1 Rain_i + \gamma w_i + e_i$, where $Rain_i$ is rainfall on election day. I vary π_1 to study the role of instrument strength while holding $\kappa = \gamma\delta$ constant so the OVB remains unchanged. I calibrate the simulations on existing studies in this literature (Gomez et al. 2007, Hansford and Gomez 2010, Fujiwara et al. 2016, Cooperman 2017); further details are on the [project’s website](#). For each value of IV strength, I create 1000 datasets of 30,000 observations and estimate both a naive OLS and a 2SLS model.

Results. Figure 5 displays, for different IV strengths, the average of statistically significant estimates scaled by the true effect size for both the IV and the naive regression model. When the instrument is strong, the IV recovers the true effect, unlike the naive regression model. Yet, as IV strength decreases, the exaggeration of statistically significant estimates increases. Even when the omitted variable bias is large, exaggeration can exceed this bias for weak instruments. IV strengths similar to those observed in the literature on which the simulations are calibrated would yield an exaggeration factor of approximately 2.9. When the only available instrument is weak, using the naive regression model might, on average, produce

9. I abstract from potential exclusion restriction violations and simulate the instrument as exogenous.

Figure 5 – Evolution of the Bias of Statistically Significant Estimates Against Strength of the Instrument in the IV Simulations



Notes: 1000 iterations. Significance level: 5%, $N = 30,000$. The blue line indicates the average bias ratio for IV statistically significant estimates. The orange line represents the same quantity for the OLS. Details on the simulation are available at this [link](#).

statistically significant estimates closer to the true effect size than the IV. Importantly for applied research, a large first-stage F -statistic does not necessarily mitigate this problem. Because $F = N \cdot R^2_{\text{partial}} / (1 - R^2_{\text{partial}})$ (Keane and Neal 2024), the F -statistic can exceed conventional thresholds even when the instrument explains a negligible share of the variation in the treatment, as long as the sample size is large enough (Staiger and Stock 1997, Bound et al. 1995). The partial R^2 , which does not grow with sample size, directly captures the share of treatment variation used for identification and is therefore a more informative diagnostic for the risk of exaggeration. This result complements findings on the limitations of first-stage F -statistics with non-iid errors (Young 2022, Lal et al. 2024).

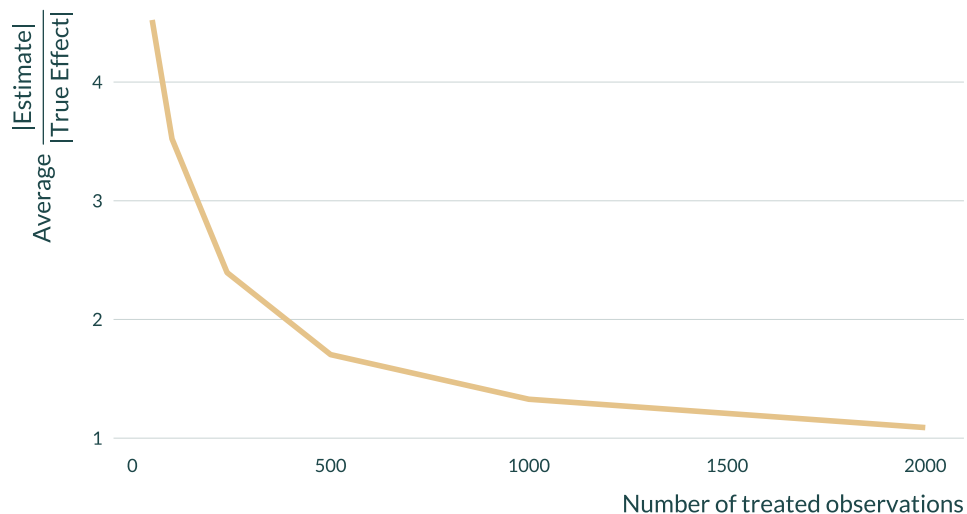
III. 4 EXOGENOUS SHOCKS

Intuition. Designs exploiting exogenous shocks achieve identification through the variation generated by those shocks. The exogeneity of the shocks is what eliminates confounding, but when the number of events or treated observations is limited, the variation available for identification is restricted. As extensively discussed in the randomized controlled trial literature, statistical power is maximized when the proportion of treated and untreated observations is balanced, and drops when one proportion approaches zero. In studies using discrete exogenous shocks, the confounding-exaggeration trade-off is thus mediated by the number of treated observations.

Case-study and simulation procedure. To illustrate this trade-off, I simulate a study of the impact of air pollution reduction on newborn weight of babies. To avoid confounding, one can exploit exogenous shocks to air pollution such as plant closures, creation of a low emission zone or of an urban toll (Currie et al. 2015, Lavaine and Neidell 2017). I simulate a typical analysis, at the zip code and monthly levels and focus on the example of toxic plant closures. I consider that the average birth weight in zip code z at time period t , bw_{zt} , depends on a zip code fixed effect ζ_z , a time fixed effect τ_t , and the treatment status T_{zt} , equal to one if a plant is closed in this period and 0 otherwise. The average birth weight in zip code z at time t is defined as follows: $bw_{zt} = \beta_0 + \beta_1 T_{zt} + \zeta_z + \tau_t + u_{zt}$. To further facilitate the identification of the effect, I assume a random treatment allocation across periods and zip codes as well as constant and homogenous effects. I only vary the proportion of observations affected by toxic plant closings.

I derive parameters values from studies from the literature (Currie et al. 2015, Lavaine and Neidell 2017, Yang and Chou 2018). The simulations mimic the design of such studies, considering a similar number of observations (100,000), distributions of variables, treatment allocation procedure and treatment effect sizes. I generate datasets with an increasing number of treated observations, varying the proportion of treated units and estimate the correct two-way fixed effects model. The simulation procedure is described in details on the [project's website](#).

Figure 6 – Evolution of Bias With the Number of Treated Observations, for Statistically Significant Estimates, in the Exogenous Shocks Simulations



Notes: 1000 iterations. Significance level: 5%, $N = 100,000$. The line indicates the average bias ratio of statistically significant estimates. Details on the simulation are available at this [link](#).

Results. Figure 6 displays the results of these simulations. As expected, exaggeration is larger for a smaller number of treated observations, keeping everything else constant. Even though the actual sample size is extremely large in this example, if the number of treated observations is small, as it can be the case in this literature, exaggeration can be substantial. For instance, one of the papers in this literature displays 240 treated observations (out of 120,000), a scale that would produce in my analogous, yet conservative, simulated setting an exaggeration ratio of about 2.4. A very large number of observations does not necessarily shield us from exaggeration.

IV NAVIGATING THE TRADE-OFF IN A STUDY

In the previous sections, I showed that using causal identification strategies induces a trade-off between avoiding confounding and exaggerating true effects. How can we, as applied researchers using observational data, arbitrate it? Since key pieces of information such as the true effect and the magnitude of confounding are inherently unknown, we cannot directly compute these biases. In this section, I examine how to gauge threats from both sides of the trade-off and, more centrally, how to characterize the variation that actually drives identification in an applied study.

IV. 1 GAUGING THE TWO SIDES OF THE TRADE-OFF

On one side of the trade-off lies the widely discussed bias caused by confounders. Although it is in essence impossible to measure, sensitivity analyses can help gauge its potential magnitude (Rosenbaum 2002, Middleton et al. 2016, Oster 2019, Cinelli and Hazlett 2020). I suggest using such analyses to assess how restrictive a causal approach must be to limit unobserved confounding to acceptable levels. In addition, given the risks of exaggeration, causal estimates should not necessarily be treated as definitive benchmarks for assessing naive estimates, nor should a discrepancy between the two be taken as sufficient evidence that the naive estimate is incorrect.

On the other side of the trade-off lies exaggeration, which emerges when statistical power is low. Like OVB, exaggeration cannot be directly measured since its computation requires knowing the true effect, which is always unknown. Power calculations can however help assess it by making assumptions about the magnitude of the true effect. In randomized controlled trials, such computations are an established requirement (Duflo et al. 2007, McConnell and Vera-Hernández 2025, Athey and Imbens 2017), yet they are rarely reported in non-experimental studies, and most causal inference textbooks devote little space to the

issue in observational settings (Angrist and Pischke 2009; 2014, Imbens and Rubin 2015, Cunningham 2021). I advocate making power more central to non-experimental analyses, even after a significant estimate has been obtained. Results from power and exaggeration calculations would be both informative and straightforward to report in robustness sections.

Prospective simulations, such as those implemented in Section III, allow evaluating power before the analysis by simulating the research design from scratch or by starting from an existing dataset with an added treatment effect (Hill 2011, Gelman 2020, Black et al. 2022).¹⁰ They allow gauging risks of exaggeration and determining how to best invest resources to avoid them: for instance increasing sample size, considering alternative identification strategies or improving measurement.

Post-analysis power calculations allow assessing whether a study’s design would have sufficient power to detect a true effect smaller than the observed estimate (Gelman and Carlin 2014, Ioannidis et al. 2017, Stommes et al. 2023). For instance, Card (1993) finds that an additional year of education causes a 13.2% increase in wage (s.e. 5.5%), using distance to a four-year college as an instrument. If the existing literature suggests that the true effect is closer to 10%, the average statistically significant estimate of Card (1993)’s design would be approximately 15% and statistical power only 44%, making the design underpowered with substantial exaggeration.¹¹ Such calculations require assumptions about the true effect size, derivable from meta-analyses, credible studies, or theoretical findings, keeping in mind that previously published estimates may themselves be exaggerated (Gelman and Carlin 2014). When such information is not available, power can be evaluated for a range of plausible smaller effect sizes.

IV. 2 ADDRESSING EXAGGERATION

In addition to these diagnostic tools, more direct solutions exist for each side of the trade-off. Causal identification strategies directly address confounding. For exaggeration, when multiple estimates of the same parameter are available, publication bias can be directly assessed and corrected. Meta-analytical methods that model the selection process recover bias-corrected estimates (Andrews and Kasy 2019). Bayesian shrinkage uses the

10. The replication material and R code I provide for Section III can be used as templates to implement simulations for most causal identification strategies, based on data generated from scratch. Similarly, I provide code and describe implementation of real-data simulations in a companion paper and in its replication material (Bagilet 2023).

11. Timm et al. (2019) and Linden (2019) offer R and Stata packages that enable running these calculations in a single command: `retrodesign(10, 5.5)`. Exaggeration is computed by drawing repeatedly from a normal distribution centered on the hypothetical effect size of 10% with a standard deviation equal to the standard error of the estimate found in the study (here 5.5%), then computing the average of the absolute value of the draws that are 1.96 standard errors away from zero.

prior distribution of effect sizes in the literature to adjust individual estimates (Zwet and Gelman 2021, Zwet and Cator 2021). Both approaches, however, require access to multiple estimates of the same parameter. They address exaggeration through the significance-filter lens, while this paper focuses on its other ingredient: imprecision. The next section introduces a complementary tool that characterizes the variation driving identification in a given study and assesses whether it is sufficient for reliable estimation, without requiring multiple estimates or assumptions about the true effect size.

More broadly, exaggeration highlights that variance remains important for the reliability of the point estimate, for reasons that go beyond mere uncertainty, even after statistical significance has been achieved. Obtaining a statistically significant estimate from an imprecise estimator should not necessarily be interpreted as a sign of “success” in getting significance despite a wide confidence interval. Rather, it may signal that this estimate lies in the tails of the distribution and thus inaccurately represents the true effect. This paper therefore calls for careful attention to the implications of design and modeling choices on estimator variance, even when large variance has not prevented statistical significance.

It is also worth considering how significant results arise in the first place when power is low. In such situations, they can be obtained either by chance or through the garden of forking paths (Simmons et al. 2011, Gelman and Loken 2013, Kasy 2021). Decision forks appear at various stages along the path of research: for instance, in data preparation, when deciding whether to include a control variable or observation in the sample, or when deciding whether to pursue research that yields non-significant results. Because of structural incentives that favor significance, the path followed may be more likely to lead to a statistically significant result. These choices most often do not reflect poor researcher practices but rather emerge from a system that treats significant results as an end goal of research.

IV. 3 IDENTIFYING THE IDENTIFYING VARIATION

IV. 3.1 A METRIC FOR IDENTIFYING OBSERVATIONS

The approaches outlined above are built on assumptions regarding the true effect size or rely on meta-analyses. To circumvent these assumptions, one can instead directly focus on the driver of the trade-off: the variation used for identification. Pinpointing the specific variation on which estimation relies can provide a direct way to gauge the risk of exaggeration.

In practice, identifying where the variation comes from is relatively straightforward in simple cases such as one-way fixed effects, but becomes more challenging in complex designs with multiple or interacting fixed effects for instance. To address this, I propose observation-specific weights that quantify each observation’s contribution to the estimation of the treat-

ment coefficient. Standard influence measures such as leverage and Cook’s distance assess influence on the entire parameter vector and are not suited for this purpose. Instead, I apply the Frisch-Waugh-Lovell theorem and compute leverage for the regression of the residualized outcome on the residualized treatment, where residuals are obtained from regressions on the full set of controls including fixed effects and control functions. This yields observation-specific weights measuring each unit’s contribution to the treatment effect estimate.

These identifying-variation weights are equivalent—up to a normalization to one—to the multiple regression weights introduced in [Aronow and Samii \(2016\)](#) and previously discussed in [Angrist and Pischke \(2009\)](#). The derivation relies on Frisch-Waugh-Lovell and is therefore native to linear regression. In instrumental variables settings such as 2SLS, the analogous identifying variation is the share of treatment variance explained by the instrument, and the weights framework adapts directly. Extensions to nonparametric estimators, including double machine learning, remain an area for further development. The estimate of the treatment coefficient in a simple linear regression can be interpreted as a weighted average of individual treatment effects. The weight ω_i of individual i is the squared difference between its treatment status T_i and the value of this treatment status as predicted by the other covariates X : $\omega_i = (T_i - \mathbb{E}[T_i|X_i])^2$. At the group level, these weights connect to well-established insights, particularly in the widely used case of fixed effects. A group’s weight, *i.e.* the sum of the individual weights within that group, corresponds to the within-group variance of the conditional treatment status. Since including fixed effects changes the type of variation used for identification from pooled to within-group variation, groups with little within variation in treatment status only marginally contribute to identification. These weights also have crucial implications at the observation level: individual observations whose treatment status is largely explained by covariates contribute little, if at all, to estimation. Variation in the explanatory variable of interest is absorbed by the controls. In that sense, the weights not only highlight the role of groups but also the uneven contribution of individual observations within them. More broadly, in settings without a natural grouping structure, observation-level weights help assess how much each unit actually contributes to identification. [Aronow and Samii \(2016\)](#) uses the multiple regression weights to argue that, when treatment effects are heterogeneous, some observations may be disproportionately represented in the estimated average effect, raising concerns about external validity. Even in the absence of heterogeneity, however, relying on observations whose treatment status is well explained by covariates can yield a very small *effective* sample size and, in turn, exaggeration. Whereas [Aronow and Samii \(2016\)](#) emphasizes the representativity of the effective sample for external validity reasons, the present paper focuses on its size, for fear of exaggeration. It also does not focus on high-weight observations but instead on the lack of contribution of low-weight ones.

More broadly, characterizing the identifying variation matters regardless of the specific concern at hand. Whether for exaggeration or for external validity, understanding which observations actually drive the estimate is essential to interpreting it. The concerns are complementary: a small effective sample raises the risk of exaggeration, while an unrepresentative one raises concerns about external validity. The `ididvar` package addresses both simultaneously, by making the structure and amount of identifying variation explicit.

IV. 3.2 A TOOL TO IDENTIFY THE IDENTIFYING VARIATION

Although the computation of these weights is relatively straightforward, it involves several steps, and their subsequent analysis and comparison are more challenging. To facilitate this process, I develop the `ididvar` package for R.¹² It provides a unified tool to compute identifying-variation weights and related contribution metrics, along with a series of visualization tools. By lowering the cost of identifying which observations drive estimation, the package aims at helping assess potential risks of exaggeration when identifying variation is limited. Unlike power calculations, these metrics can be computed without assumptions about the true effect size.

Most functions are high-level and easy to use, requiring only the regression output and the variable of interest. A first set of functions computes and visualizes weights, including their distribution across space, groups, or time. This enables users to detect low-weight observations or clusters that contribute little to identification. Because the distribution of weights is often heterogeneous, building meaningful visualization can be challenging. To address this, I propose a discretized and logged scale that compares each weight to the average weight, $1/n$, where n is the number of observations or groups. This representation clarifies visualization and ensures comparability across specifications and studies.

To illustrate this feature of the package, I consider a simple example using a country-level panel (Bryan 2025, restricted to Africa and Europe for clarity) and estimate the within-country relationship between GDP per capita and life expectancy: $\log(\text{PerCapitaGDP}_{ct}) = \alpha_c + \beta \text{LifeExp}_{ct} + u_{ct}$. Figure 7 presents visualizations produced by the package. Panel A displays the identifying-variation weights for each observation,¹³ while Panel B shows their cumulative distribution. The distribution is highly heterogeneous: many observations contribute little to identification, particularly those from mid-periods and European countries.

12. More information on the package is available on [the package’s website](#). This package can be extended to accommodate more specific settings. Readers are encouraged to suggest improvements through [GitHub issues](#) or [pull requests](#).

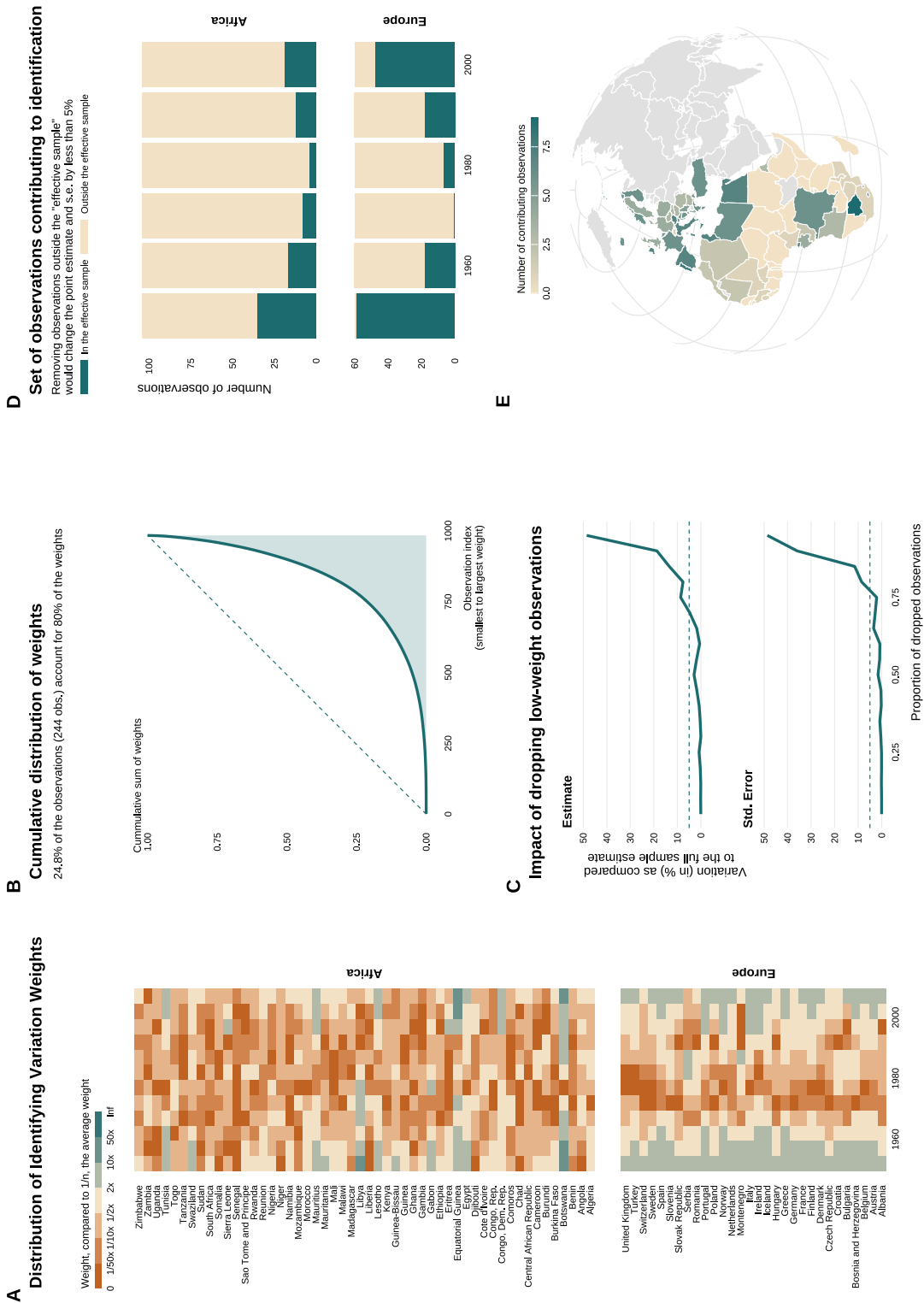
13. This graph is essentially generated in one line of code: `idid_viz_weights(reg, "l_gdpPercap", year, country)` where `reg` is the output of the regression of interest, `l_gdpPercap` the variable of interest and `year` and `country` specify the variables to be displayed on the x and y -axes respectively.

Weights alone, however, do not indicate which observations actually contribute to identification. A second set of functions identifies the *effective sample*: the smallest subset of observations that cannot be removed without altering the point estimate or standard error of the parameter of interest by more than a chosen margin, e.g., 5%. The procedure iteratively re-estimates the model, removing low-weight observations following an approach similar to influence-function methods. Reporting the effective sample size helps interpret results in light of the sample that actually drives identification, much as one would interpret the nominal sample size.

In the illustrative example, Panel C of Figure 7 shows how point estimates and standard errors evolve as low-weight observations are sequentially dropped. 70% of the nominal sample can be dropped without affecting estimates and standard errors by more than 5%. The effective sample is thus about four times smaller than the nominal sample ($n_{\text{nominal}} = 984$ and $n_{\text{effective}} = 246$). Results should be interpreted accordingly. Panels D and E characterize this effective sample: many observations from Africa and mid-periods do not contribute to identification, implying that the effective sample, in addition to being small, is not representative of the full dataset.

The package is flexible and can be applied to a wide range of identification strategies and estimation methods. It enables users to compare specifications and explore how the inclusion of different sets of fixed effects or controls affects the effective sample. More broadly, by making the structure and amount of identifying variation explicit, the package provides a concrete way to assess where a study lies along the confounding-exaggeration trade-off. In doing so, it aims to help users evaluate whether the credibility gained from additional controls comes at the cost of a loss of identifying variation and statistical power.

Figure 7 – Illustrative Visualizations and Analyses Produced with ididvar



V CONCLUSION

The foundational component of causal inference is that, to identify causal effects, it limits the variation used for estimation to the exogenous part. This is what makes causal methods credible, but I show that it can also reduce statistical power and increase the risk of exaggeration. Exaggeration of published estimates arises when two conditions are met: selection on significance, which is well documented in the economics literature (Rosenthal 1979, Andrews and Kasy 2019, Vivaldi 2019, Abadie 2020, Brodeur et al. 2020), and low statistical power (Gelman and Tuerlinckx 2000, Ioannidis 2008, Gelman and Carlin 2014). I highlight that causal identification strategies can lead to low power when they leverage only a small amount of variation for identification. Through theory and calibrated simulations, I show that the exaggeration bias can exceed the confounding bias the strategy aims to eliminate. Causal exaggeration is driven by the variation that remains for identification. To help diagnose it in an applied study, I introduce a tool to identify the identifying variation: the observations and sources of variation that actually drive the causal estimate. Systematically reporting statistical power and examining the variation actually used for identification, including by describing the effective sample and reporting its size alongside the nominal one, could help navigate this confounding-exaggeration trade-off.

This paper has primarily taken the perspective of individual studies, in particular when focusing on one of the two ingredients of exaggeration, low power. But the other ingredient for exaggeration, selection on significance, is partly structural. As shown in the simulations, if estimates were not filtered by their statistical significance, even underpowered studies would on average recover the true effect, as long as the estimator is unbiased. Many researchers advocate moving away from statistical significance as a measure of study quality (McShane et al. 2019, Abadie 2020). To be effective, this change should be paired with increased replication (Christensen and Miguel 2018): replications, even of underpowered studies, would eventually enable constructing the distribution of the causal estimand of interest, and meta-analyses would reduce uncertainty around its true value by pooling estimates (Hernán 2022). Interpreting confidence intervals as compatibility intervals, with their width indicating a range of effect sizes compatible with the data, would further limit inflation from selection on significance (Amrhein et al. 2019, Romer 2020).

REFERENCES

- Abadie, A. (2020). Statistical Nonsignificance in Empirical Economics. *American Economic Review: Insights*, 2(2):193–208.
- Amrhein, V., Trafimow, D., and Greenland, S. (2019). Inferential Statistics as Descriptive Statistics: There Is No Replication Crisis if We Don’t Expect Replication. *The American Statistician*, 73(sup1):262–270.
- Andrews, I. and Kasy, M. (2019). Identification of and Correction for Publication Bias. *American Economic Review*, 109(8):2766–2794.
- Angrist, J. D. and Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist’s Companion*. Princeton University Press, first edition edition.
- Angrist, J. D. and Pischke, J.-S. (2014). *Mastering ’Metrics: The Path from Cause to Effect*. Princeton University Press.
- Arel-Bundock, V., Briggs, R. C., Doucouliagos, H., Aviña, M. M., and Stanley, T. D. (2024). Quantitative Political Science Research Is Greatly Underpowered. *The Journal of Politics*, 88(1):36–46.
- Aronow, P. M. and Samii, C. (2016). Does Regression Produce Representative Estimates of Causal Effects? *American Journal of Political Science*, 60(1):250–267.
- Athey, S. and Imbens, G. (2017). The Econometrics of Randomized Experiments. In *Handbook of Economic Field Experiments*, volume 1, pages 73–140. North-Holland.
- Bagilet, V. (2023). Accurate Estimation of Small Effects: Illustration Through Air Pollution and Health.
- Black, B., Hollingsworth, A., Nunes, L., and Simon, K. (2022). Simulated power analyses for observational studies: An application to the Affordable Care Act Medicaid expansion. *Journal of Public Economics*, 213:104713.
- Bound, J., Jaeger, D. A., and Baker, R. M. (1995). Problems with Instrumental Variables Estimation When the Correlation Between the Instruments and the Endogeneous Explanatory Variable is Weak. *Journal of the American Statistical Association*, 90(430):443–450.
- Brodeur, A., Cook, N., and Heyes, A. (2020). Methods Matter: P-Hacking and Publication Bias in Causal Analysis in Economics. *American Economic Review*, 110(11):3634–3660.
- Brodeur, A., Lé, M., Sangnier, M., and Zylberberg, Y. (2016). Star Wars: The Empirics Strike Back. *American Economic Journal: Applied Economics*, 8(1):1–32.
- Bryan, J. (2025). Gapminder: Data from Gapminder.
- Camerer, C. F., Dreber, A., Forsell, E., Ho, T.-H., Huber, J., Johannesson, M., Kirchler, M., Almenberg, J., Altmejd, A., Chan, T., Heikensten, E., Holzmeister, F., Imai, T., Isaksson, S., Nave, G., Pfeiffer, T., Razen, M., and Wu, H. (2016). Evaluating replicability of laboratory experiments in economics. *Science*, 351(6280):1433–1436.
- Card, D. (1993). Using Geographic Variation in College Proximity to Estimate the Return to Schooling. Working Paper 4483, National Bureau of Economic Research.

- Christensen, G. and Miguel, E. (2018). Transparency, Reproducibility, and the Credibility of Economics Research. *Journal of Economic Literature*, 56(3):920–980.
- Cinelli, C. and Hazlett, C. (2020). Making sense of sensitivity: Extending omitted variable bias. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 82(1):39–67.
- Cooperman, A. D. (2017). Randomization Inference with Rainfall Data: Using Historical Weather Patterns for Variance Estimation. *Political Analysis*, 25(3):277–288.
- Cunningham, S. (2021). *Causal Inference: The Mixtape*. Yale University Press.
- Currie, J., Davis, L., Greenstone, M., and Walker, R. (2015). Environmental Health Risks and Housing Values: Evidence from 1,600 Toxic Plant Openings and Closings. *American Economic Review*, 105(2):678–709.
- Deaton, A. and Cartwright, N. (2018). Understanding and misunderstanding randomized controlled trials. *Social Science & Medicine*, 210:2–21.
- DellaVigna, S. and Linos, E. (2022). RCTs to Scale: Comprehensive Evidence From Two Nudge Units. *Econometrica*, 90(1):81–116.
- Duflo, E., Glennerster, R., and Kremer, M. (2007). Using Randomization in Development Economics Research: A Toolkit. In Schultz, T. P. and Strauss, J. A., editors, *Handbook of Development Economics*, volume 4, pages 3895–3962. Elsevier.
- Ferraro, P. J. and Shukla, P. (2020). Feature—Is a Replicability Crisis on the Horizon for Environmental and Resource Economics? *Review of Environmental Economics and Policy*, 14(2):339–351.
- Fujiwara, T., Meng, K., and Vogl, T. (2016). Habit Formation in Voting: Evidence from Rainy Elections. *American Economic Journal: Applied Economics*, 8(4):160–188.
- Gelman, A. (2020). *Regression and Other Stories*. Cambridge University Press, Cambridge New York, NY Port Melbourne, VIC New Delhi Singapore.
- Gelman, A. and Carlin, J. (2014). Beyond Power Calculations: Assessing Type S (Sign) and Type M (Magnitude) Errors. *Perspectives on Psychological Science*, 9(6):641–651.
- Gelman, A. and Loken, E. (2013). The garden of forking paths: Why multiple comparisons can be a problem, even when there is no “fishing expedition” or “p-hacking” and the research hypothesis was posited ahead of time.
- Gelman, A. and Tuerlinckx, F. (2000). Type S error rates for classical and Bayesian single and multiple comparison procedures. *Computational Statistics*, 15(3):373–390.
- Gomez, B. T., Hansford, T. G., and Krause, G. A. (2007). The Republicans Should Pray for Rain: Weather, Turnout, and Voting in U.S. Presidential Elections. *The Journal of Politics*, 69(3):649–663.
- Hansford, T. G. and Gomez, B. T. (2010). Estimating the Electoral Effects of Voter Turnout. *American Political Science Review*, 104(2):268–288.
- Hernán, M. A. (2022). Causal analyses of existing databases: No power calculations required. *Journal of Clinical Epidemiology*, 144:203–205.

- Hernán, M. A. and Robins, J. M. (2020). *Causal Inference: What If*. Boca Raton: Chapman & Hall/CRC.
- Hill, J. L. (2011). Bayesian Nonparametric Modeling for Causal Inference. *Journal of Computational and Graphical Statistics*, 20(1):217–240.
- Imbens, G. and Kalyanaraman, K. (2012). Optimal Bandwidth Choice for the Regression Discontinuity Estimator. *The Review of Economic Studies*, 79(3):933–959.
- Imbens, G. W. and Rubin, D. B. (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. Cambridge University Press.
- Ioannidis, J. P. A. (2008). Why Most Discovered True Associations Are Inflated. *Epidemiology*, 19(5):640–648.
- Ioannidis, J. P. A., Stanley, T. D., and Doucouliagos, H. (2017). The Power of Bias in Economics Research. *The Economic Journal*, 127(605):F236–F265.
- Jacob, B. A. and Lefgren, L. (2004). Remedial Education and Student Achievement: A Regression-Discontinuity Analysis. *The Review of Economics and Statistics*, 86(1):226–244.
- Kasy, M. (2021). Of Forking Paths and Tied Hands: Selective Publication of Findings, and What Economists Should Do about It. *Journal of Economic Perspectives*, 35(3):175–192.
- Keane, M. P. and Neal, T. (2024). A Practical Guide to Weak Instruments. *Annual Review of Economics*, 16(Volume 16, 2024):185–212.
- Kraft, M. A. (2020). Interpreting Effect Sizes of Education Interventions. *Educational Researcher*, 49(4):241–253.
- Lai, W., Qiu, Y., Tang, Q., Xi, C., and Zhang, P. (2023). The Effects of Temperature on Labor Productivity. *Annual Review of Resource Economics*, 15:213–232.
- Lal, A., Lockhart, M., Xu, Y., and Zu, Z. (2024). How Much Should We Trust Instrumental Variable Estimates in Political Science? Practical Advice Based on 67 Replicated Studies. *Political Analysis*, pages 1–20.
- Lavaine, E. and Neidell, M. (2017). Energy Production and Health Externalities: Evidence from Oil Refinery Strikes in France. *Journal of the Association of Environmental and Resource Economists*, 4(2):447–477.
- Linden, A. (2019). RETRODESIGN: Stata module to compute type-S (Sign) and type-M (Magnitude) errors. *Statistical Software Components*.
- LoPalo, M. (2023). Temperature, Worker Productivity, and Adaptation: Evidence from Survey Data Production. *American Economic Journal: Applied Economics*, 15(1):192–229.
- Lu, J., Qiu, Y., and Deng, A. (2019). A note on Type S/M errors in hypothesis testing. *British Journal of Mathematical and Statistical Psychology*, 72(1):1–17.
- McConnell, B. and Vera-Hernández, M. (2025). Going beyond simple sample size calculations: A practitioner’s guide. *Fiscal Studies*, 46(3):323–348.
- McShane, B. B., Gal, D., Gelman, A., Robert, C., and Tackett, J. L. (2019). Abandon

- Statistical Significance. *The American Statistician*, 73(sup1):235–245.
- Middleton, J. A., Scott, M. A., Diakow, R., and Hill, J. L. (2016). Bias Amplification and Bias Unmasking. *Political Analysis*, 24(3):307–323.
- Oster, E. (2019). Unobservable Selection and Coefficient Stability: Theory and Evidence. *Journal of Business & Economic Statistics*, 37(2):187–204.
- Ravallion, M. (2020). Should the Randomistas (Continue to) Rule? In Bédécarrats, F., Guérin, I., and Roubaud, F., editors, *Randomized Control Trials in the Field of Development: A Critical Perspective*. Oxford University Press.
- Romer, D. (2020). In Praise of Confidence Intervals. *AEA Papers and Proceedings*, 110:55–60.
- Rosenbaum, P. R. (2002). *Observational Studies*. Springer New York, New York, NY.
- Rosenthal, R. (1979). The file drawer problem and tolerance for null results. *Psychological Bulletin*, 86(3):638–641.
- Schell, T. L., Griffin, B. A., and Morral, A. R. (2018). Evaluating Methods to Estimate the Effect of State Laws on Firearm Deaths: A Simulation Study. Technical report, RAND Corporation.
- Simmons, J. P., Nelson, L. D., and Simonsohn, U. (2011). False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant. *Psychological Science*, 22(11):1359–1366.
- Somanathan, E., Somanathan, R., Sudarshan, A., and Tewari, M. (2021). The Impact of Temperature on Productivity and Labor Supply: Evidence from Indian Manufacturing. *Journal of Political Economy*, 129(6):1797–1827.
- Staiger, D. and Stock, J. H. (1997). Instrumental Variables Regression with Weak Instruments. *Econometrica*, 65(3):557–586.
- Stevens, A. (2017). Temperature, Wages, and Agricultural Labor Productivity.
- Stommes, D., Aronow, P. M., and Sävje, F. (2023). On the reliability of published findings using the regression discontinuity design in political science. *Research & Politics*, 10(2).
- Thistlethwaite, D. L. and Campbell, D. T. (1960). Regression-discontinuity analysis: An alternative to the ex post facto experiment. *Journal of Educational Psychology*, 51(6):309–317.
- Timm, A., Gelman, A., and Carlin, J. (2019). Retrodesign: Tools for Type S (Sign) and Type M (Magnitude) Errors.
- Vivalt, E. (2019). Specification Searching and Significance Inflation Across Time, Methods and Disciplines. *Oxford Bulletin of Economics and Statistics*, 81(4):797–816.
- Vu, P. (2025). Do Standard Error Corrections Exacerbate Publication Bias?
- Yang, M. and Chou, S.-Y. (2018). The impact of environmental regulation on fetal health: Evidence from the shutdown of a coal-fired power plant located upwind of New Jersey. *Journal of Environmental Economics and Management*, 90:269–293.
- Young, A. (2022). Consistency without Inference: Instrumental Variables in Practical Ap-

plication. *European Economic Review*, 147:104112.

Zwet, E. and Gelman, A. (2021). A Proposal for Informative Default Priors Scaled by the Standard Error of Estimates. *The American Statistician*, pages 1–9.

Zwet, E. W. and Cator, E. A. (2021). The significance filter, the winner’s curse and the need to shrink. *Statistica Neerlandica*, 75(4):437–452.

A MATHEMATICAL PROOFS

I. 1 VARIATION OF THE EXAGGERATION RATIO (LEMMA 1)

Proof. Lu et al. (2019) and Zwet and Cator (2021) showed this in the case of $b = 0$. To extend it to the biased case, consider $E_b = \frac{\mathbb{E} \left[|\hat{\beta}_b| \mid \beta_1, \sigma, |\hat{\beta}_b| > z_\alpha \sigma \right]}{|\beta_1|}$ the exaggeration ratio of interest. Note that, since $\hat{\beta}_b$ is an unbiased estimator of $\beta_1 + b$, $\tilde{E}_b = \frac{\mathbb{E} \left[|\hat{\beta}_b| \mid \beta_1, \sigma, |\hat{\beta}_b| > z_\alpha \sigma \right]}{|\beta_1 + b|}$ has the properties described in the lemma. Now, considering that $E_b = \left| \frac{\beta_1 + b}{\beta_1} \right| \tilde{E}_b$ shows the properties when β_1 and b have the same sign. $\square$

I. 2 ASYMPTOTIC DISTRIBUTION OF $\hat{\beta}_{\text{OVB}}$ (LEMMA 2)

For readability, let us introduce the usual vector notation such that for instance $y = (y_1, \dots, y_n)'$ and set $\beta = (\beta_0, \beta_1)'$ and $x_i = (1, x_i)'$. I also use capital letters to denote matrices (for instance $X = (x_1', \dots, x_n')'$).

Proof. Since, we do not observe w , we consider the projection of y on X only:

$$y = X\beta_{\text{OVB}} + u_{\text{OVB}} \quad (5)$$

where by definition of the projection, $\mathbb{E}[X'u_{\text{OVB}}] = 0$.

We first compute the bias of the estimator. From equation 5 we get:

$$\begin{aligned} X'y &= X'X\beta_{\text{OVB}} + X'u_{\text{OVB}} \\ \Rightarrow \mathbb{E}[X'y] &= \underbrace{\mathbb{E}[X'X]}_{\text{pos. def.}} \beta_{\text{OVB}} + \underbrace{\mathbb{E}[X'u_{\text{OVB}}]}_0 \\ \Leftrightarrow \beta_{\text{OVB}} &= \mathbb{E}[X'X]^{-1} \mathbb{E}[X'(X\beta + \delta w + u)] \quad \text{cf eq. 2} \\ \Leftrightarrow \beta_{\text{OVB}} &= \beta + \mathbb{E}[X'X]^{-1} \mathbb{E}[X'w] \delta \end{aligned} \quad (6)$$

We then compute the asymptotic distribution. We can write:

$$\sqrt{n}(\hat{\beta}_{\text{OVB}} - \beta_{\text{OVB}}) = \left(\frac{1}{n} \sum_{i=1}^n x_i x_i' \right)^{-1} \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n x_i u_{\text{OVB},i} \right)$$

Applying the Weak Law of Large Numbers (WLLN), the Central Limit Theorem (CLT) and Slutsky's theorem yields:

$$\sqrt{n}(\hat{\beta}_{\text{OVb}} - \beta_{\text{OVb}}) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathbf{x}_i \mathbf{x}_i']^{-1} \mathbb{E}[\mathbf{x}_i \mathbf{x}_i' u_{\text{OVb},i}^2] \mathbb{E}[\mathbf{x}_i \mathbf{x}_i']^{-1}) \quad (7)$$

We are interested in the second component of $\hat{\beta}_{\text{OVb}}$. To retrieve it we need to compute $\mathbb{E}[\mathbf{x}_i \mathbf{x}_i']^{-1}$, $\mathbb{E}[\mathbf{x}_i w_i]$ and $\mathbb{E}[\mathbf{x}_i \mathbf{x}_i' u_{\text{OVb},i}^2]$.

$$\begin{aligned} \mathbb{E}[\mathbf{x}_i \mathbf{x}_i'] &= \mathbb{E} \begin{bmatrix} 1 & x_i \\ x_i & x_i^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_x \\ \mu_x & \sigma_x^2 + \mu_x^2 \end{bmatrix} \Rightarrow \mathbb{E}[\mathbf{x}_i \mathbf{x}_i']^{-1} = \frac{1}{\sigma_x^2} \begin{bmatrix} \sigma_x^2 + \mu_x^2 & -\mu_x \\ -\mu_x & 1 \end{bmatrix} \\ \mathbb{E}[\mathbf{x}_i w_i] &= \mathbb{E} \begin{bmatrix} w_i \\ x_i w_i \end{bmatrix} = \begin{bmatrix} 0 \\ \underbrace{\mathbb{E}[x_i] \mathbb{E}[w_i] + \text{cov}(x_i, w_i)}_0 \end{bmatrix} = \begin{bmatrix} 0 \\ \underbrace{\gamma \text{var}(w_i) + \underbrace{\text{cov}(\epsilon_i, w_i)}_0}_{\sigma_w^2} \end{bmatrix} = \begin{bmatrix} 0 \\ \gamma \sigma_w^2 \end{bmatrix} \\ &\Rightarrow \mathbb{E}[\mathbf{x}_i \mathbf{x}_i']^{-1} \mathbb{E}[\mathbf{x}_i w_i] = \frac{\gamma \sigma_w^2}{\sigma_x^2} \begin{bmatrix} -\mu_x \\ 1 \end{bmatrix} \end{aligned} \quad (8)$$

Note that $\mathbb{E}[\mathbf{x}_i \mathbf{x}_i' u_{\text{OVb},i}^2] \stackrel{\text{LIE}}{=} \mathbb{E}[\mathbf{x}_i \mathbf{x}_i' \mathbb{E}[u_{\text{OVb},i}^2 | \mathbf{x}_i]]$. We thus first compute $\mathbb{E}[u_{\text{OVb},i}^2 | \mathbf{x}_i]$, noting:

$$\begin{aligned} u_{\text{OVb},i} &= y_i - \mathbf{x}_i' \beta_{\text{OVb}} \\ &= \delta w_i + u_i + \mathbf{x}_i' (\beta - \beta_{\text{OVb}}) \\ &= \delta w_i + u_i - \underbrace{\mathbf{x}_i' \mathbb{E}[\mathbf{x}_i \mathbf{x}_i']^{-1} \mathbb{E}[\mathbf{x}_i w_i]}_{\text{projection of } w_i \text{ on } x_i} \delta \\ &= u_i + \delta \underbrace{\left(w_i - \frac{\gamma \sigma_w^2}{\sigma_x^2} (x_i - \mu_x) \right)}_{\text{part of } w_i \text{ orthogonal to } x_i} \\ &= u_i + \delta w_i^\perp \end{aligned} \quad \text{where } w_i^\perp = w_i - \frac{\gamma \sigma_w^2}{\sigma_x^2} (x_i - \mu_x)$$

And thus,

$$\begin{aligned}
\mathbb{E}[u_{\text{OVB},i}^2 | \mathbf{x}_i] &= \mathbb{E}[(u_i + \delta w_i^\perp)^2 | x_i] \\
&= \mathbb{E}[u_i^2 | x_i] + 2\delta \mathbb{E}[u_i w_i^\perp | x_i] + \delta^2 \mathbb{E}[(w_i^\perp)^2 | x_i] \\
&= \sigma_u^2 + 2\delta (\mathbb{E}[u_i w_i | x_i] - \underbrace{\frac{\gamma \sigma_w^2}{\sigma_x^2} (x_i - \mu_x) \mathbb{E}[u_i | x_i]}_0) + \delta^2 \mathbb{E}[(w_i^\perp)^2 | x_i] \\
&\stackrel{\text{LIE}}{=} \sigma_u^2 + 2\delta \underbrace{\mathbb{E}[w_i \mathbb{E}[u_i | x_i, w_i] | x_i]}_0 + \delta^2 \mathbb{E}[(w_i^\perp)^2 | x_i] \\
&= \sigma_u^2 + \delta^2 \mathbb{E}[(w_i^\perp)^2 | x_i]
\end{aligned}$$

Notice that, by the law of total variance, $\mathbb{E}[(w_i^\perp)^2 | x_i] = \text{Var}(w_i^\perp | x_i) + \mathbb{E}[w_i^\perp | x_i]^2$. Now, since $w_i^\perp$ is the component of w_i that is orthogonal to x_i and by the projection interpretation of the conditional variance, $\mathbb{E}[w_i^\perp | x_i] = 0$. And thus, since by assumption $\text{Var}(w_i^\perp | x_i) = \text{Var}(w_i^\perp)$,

$$\begin{aligned}
\mathbb{E}[(w_i^\perp)^2 | x_i] &= \text{Var}(w_i^\perp | x_i) \\
&= \text{Var}(w_i^\perp) \\
&= \mathbb{E}[(w_i^\perp)^2] - \mathbb{E}[w_i^\perp]^2 \\
&= \mathbb{E} \left[\left(w_i - \frac{\gamma \sigma_w^2}{\sigma_x^2} (x_i - \mu_x) \right)^2 \right] - \left(\underbrace{\mathbb{E}[w_i]}_0 + \frac{\gamma \sigma_w^2}{\sigma_x^2} \underbrace{\mathbb{E}[x_i - \mu_x]}_0 \right)^2 \\
&= \underbrace{\mathbb{E}[w_i^2]}_{\sigma_w^2} - 2 \frac{\gamma \sigma_w^2}{\sigma_x^2} \left(\underbrace{\mathbb{E}[x_i w_i]}_{\gamma \sigma_w^2} - \mu_x \underbrace{\mathbb{E}[w_i]}_0 \right) + \frac{\gamma^2 \sigma_w^4}{\sigma_x^4} \underbrace{\mathbb{E}[(x_i - \mu_x)^2]}_{\sigma_x^2} \\
&= \sigma_w^2 \left(1 - \frac{\gamma^2 \sigma_w^2}{\sigma_x^2} \right)
\end{aligned}$$

Note that this variance is well defined (positive) only if $\sigma_x^2 \geq \gamma^2 \sigma_w^2$. Under this condition,

$$\mathbb{E}[u_{\text{OVB},i}^2 | \mathbf{x}_i] = \sigma_u^2 + \delta^2 \sigma_w^2 \left(1 - \frac{\gamma^2 \sigma_w^2}{\sigma_x^2} \right) \quad (9)$$

Thus, under our set of assumptions, $\mathbb{E}[u_{\text{OVB},i}^2 | \mathbf{x}_i]$ does not depend on x_i and $\mathbb{E}[u_{\text{OVB},i}^2 | \mathbf{x}_i] = \mathbb{E}[u_{\text{OVB},i}^2]$. We denote this quantity $\sigma_{u_{\text{OVB}}}^2$.

We can now compute the variance of the estimator $\hat{\beta}_{\text{OVB}}$, noting that $\mathbb{E}[\mathbf{x}_i \mathbf{x}_i' u_{\text{OVB},i}^2] = \mathbb{E}[\mathbf{x}_i \mathbf{x}_i' \mathbb{E}[u_{\text{OVB},i}^2 | \mathbf{x}_i]] = \mathbb{E}[\mathbf{x}_i \mathbf{x}_i' \sigma_{u_{\text{OVB}}}^2] = \sigma_{u_{\text{OVB}}}^2 \mathbb{E}[\mathbf{x}_i \mathbf{x}_i']$. And thus $\mathbb{E}[\mathbf{x}_i \mathbf{x}_i']^{-1} \mathbb{E}[\mathbf{x}_i \mathbf{x}_i' u_{\text{OVB},i}^2] \mathbb{E}[\mathbf{x}_i \mathbf{x}_i']^{-1} = \sigma_{u_{\text{OVB}}}^2 \mathbb{E}[\mathbf{x}_i \mathbf{x}_i']$.

Plugin this and equation 8 into equation 7, we get, for $\hat{\beta}_{\text{OVB}}$, the second component of $\hat{\beta}_{\text{OVB}}$:

$$\sqrt{n}(\hat{\beta}_{\text{OVB}} - \beta_1) \xrightarrow{d} \mathcal{N}\left(\frac{\delta\gamma\sigma_w^2}{\sigma_x^2}, \frac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{\sigma_x^2}\right)$$

Then, noting that $\rho_{xw} = \text{corr}(x, w) = \frac{\text{cov}(\mu_x + \gamma w + \epsilon, w)}{\sigma_x \sigma_w} = \frac{\gamma\sigma_w}{\sigma_x}$, we have:

$$\sigma_{\text{OVB}}^2 = \text{avar}\left(\hat{\beta}_{\text{OVB}}\right) = \frac{\sigma_u^2 + \delta^2\sigma_w^2(1 - \rho_{xw}^2)}{\sigma_x^2}$$

□

I. 3 ASYMPTOTIC DISTRIBUTION OF $\hat{\beta}_{\text{CTRL}}$ (LEMMA 3)

Proof. The proof is the well know proof of the asymptotic distribution of the OLS. I simply compute $\mathbb{E}[x_{w,i}x'_{w,i}]^{-1}$ to retrieve the variance of the parameter of interest β_{CTRL} . We know that we have:

$$\sqrt{n}(\hat{\beta}_{\text{CTRL}} - \beta_{\text{CTRL}}) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[x_{w,i}x'_{w,i}]^{-1}\sigma_u^2)$$

We are interested in the second component of $\hat{\beta}_{\text{CTRL}}$. To retrieve it we need to compute $\mathbb{E}[x_{w,i}x'_{w,i}]^{-1}$.

$$\mathbb{E}[x_{w,i}x'_{w,i}] = \mathbb{E}\begin{bmatrix} 1 & x_i & w \\ x_i & x_i^2 & x_i w_i \\ w_i & x_i w_i & w_i^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_x & 0 \\ \mu_x & \sigma_x^2 + \mu_x^2 & \gamma\sigma_w^2 \\ 0 & \gamma\sigma_w^2 & \sigma_w^2 \end{bmatrix}$$

Note that we have $\mathbb{E}[x_i w_i] = \mathbb{E}[x_i] \underbrace{\mathbb{E}[w_i]}_0 + \text{cov}(x_i, w_i) = \underbrace{\gamma \text{var}(w_i)}_{\sigma_w^2} + \underbrace{\text{cov}(\epsilon_i, w_i)}_0 = \gamma\sigma_w^2$.

Now, $\mathbb{E}[x_{w,i}x'_{w,i}]^{-1} = \frac{1}{\det(\mathbb{E}[x_{w,i}x'_{w,i}])} {}^t\text{C}$ with C the comatrix of $\mathbb{E}[x_{w,i}x'_{w,i}]$. We have:

$$\det(\mathbb{E}[x_{w,i}x'_{w,i}]) = (\sigma_x^2 + \mu_x^2)\sigma_w^2 - \sigma_w^2\mu_x^2 - \gamma^2\sigma_w^4 = \sigma_w^2(\sigma_x^2 - \gamma^2\sigma_w^2)$$

and the “central” component of C, σ_w^2 . Thus the central component of interest of $\mathbb{E}[x_{w,i}x'_{w,i}]^{-1}$ is $\frac{1}{\sigma_x^2 - \gamma^2\sigma_w^2}$. Therefore, for $\hat{\beta}_{\text{CTRL}}$, the second component of $\hat{\beta}_{\text{CTRL}}$, we have:

$$\sqrt{n}(\hat{\beta}_{\text{CTRL}} - \beta_1) \xrightarrow{d} \mathcal{N}\left(0, \frac{\sigma_u^2}{\sigma_x^2 - \gamma^2\sigma_w^2}\right) \quad (10)$$

Then, noting that $\rho_{xw} = \text{corr}(x, w) = \frac{\text{cov}(\mu_x + \gamma w + \epsilon, w)}{\sigma_x \sigma_w} = \frac{\gamma \sigma_w}{\sigma_x}$, we have:

$$\sigma_{\text{CTRL}}^2 = \frac{\sigma_u^2}{\sigma_x^2(1 - \rho_{xw}^2)}$$

□

I. 4 ASYMPTOTIC DISTRIBUTION OF $\hat{\beta}_{\text{IV}}$ (LEMMA 4)

Proof. Since $u_{\text{IV}} = u_{\text{OVB}} = \delta w + u$, we have $\sigma_{u_{\text{IV}}}^2 = \sigma_u^2 + \delta^2 \sigma_w^2$. Thus, the usual asymptotic distribution of the IV gives:

$$\sqrt{n}(\hat{\beta}_{\text{IV}} - \beta) \xrightarrow{d} \mathcal{N}\left(0, (\sigma_u^2 + \delta^2 \sigma_w^2) \mathbb{E}[z_i x_i']^{-1} \mathbb{E}[z_i z_i'] (\mathbb{E}[z_i x_i']^{-1})'\right)$$

Note that, when the instruments are weak, the convergence of the sampling distribution to this asymptotic normal distribution does not hold. However, the resulting distribution having fatter tails than the normal and significant estimates from underpowered studies being located in the tails of the distribution, the normal distribution will yield a lower bound for the exaggeration.

We are interested in the second component of $\hat{\beta}_{\text{IV}}$. To retrieve it we need to compute $\mathbb{E}[z_i z_i']$, $\mathbb{E}[x_i z_i']^{-1}$ and its transpose.

$$\begin{aligned} \mathbb{E}[z_i z_i'] &= \mathbb{E} \begin{bmatrix} 1 & z_i \\ z_i & z_i^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_z \\ \mu_z & \sigma_z^2 + \mu_z^2 \end{bmatrix} \\ \mathbb{E}[z_i x_i'] &= \begin{bmatrix} 1 & \mathbb{E}[x_i] \\ \mathbb{E}[z_i] & \mathbb{E}[z_i x_i] \end{bmatrix} = \begin{bmatrix} 1 & \pi_0 + \pi_1 \mathbb{E}[z_i] + \gamma \underbrace{\mathbb{E}[w_i]}_0 + \underbrace{\mathbb{E}[e_i]}_0 \\ \mu_z & \pi_0 \mathbb{E}[z_i] + \pi_1 \mathbb{E}[z_i^2] + \gamma \underbrace{\mathbb{E}[z_i w_i]}_0 + \underbrace{\mathbb{E}[z_i e_i]}_0 \end{bmatrix} = \begin{bmatrix} 1 & \pi_0 + \pi_1 \mu_z \\ \mu_z & \pi_0 \mu_z + \pi_1 (\sigma_z^2 + \mu_z^2) \end{bmatrix} \\ \Rightarrow \mathbb{E}[z_i x_i']^{-1} &= \frac{1}{\pi_1 \sigma_z^2} \begin{bmatrix} \pi_0 \mu_z + \pi_1 (\sigma_z^2 + \mu_z^2) & -\pi_0 - \pi_1 \mu_z \\ -\mu_z & 1 \end{bmatrix} \end{aligned}$$

Thus,

$$\mathbb{E}[z_i x_i']^{-1} \mathbb{E}[z_i z_i'] (\mathbb{E}[z_i x_i']^{-1})' = \frac{1}{\pi_1 \sigma_z^2} \begin{bmatrix} 2\pi_0 \mu_z + \pi_1 (\sigma_z^2 + \mu_z^2) + \frac{\pi_0^2}{\pi_1} & -\mu_z - \frac{\pi_0}{\pi_1} \\ -\mu_z - \frac{\pi_0}{\pi_1} & \frac{1}{\pi_1} \end{bmatrix}$$

And so, for $\hat{\beta}_{\text{IV}}$, the second component of $\hat{\beta}_{\text{IV}}$, we have:

$$\sqrt{n} \left(\hat{\beta}_{\text{IV}} - \beta_1 \right) \xrightarrow{d} \mathcal{N} \left(0, \frac{\sigma_u^2 + \delta^2 \sigma_w^2}{\pi_1^2 \sigma_z^2} \right) \quad (11)$$

Now, since $\rho_{xz} = \text{corr}(x_i, z_i) = \frac{\text{cov}(\pi_0 + \pi_1 z_i + \gamma w_i + e_i, z_i)}{\sigma_x \sigma_z} = \pi_1 \frac{\sigma_z}{\sigma_x}$,

$$\sqrt{n} \left(\hat{\beta}_{\text{IV}} - \beta_1 \right) \xrightarrow{d} \mathcal{N} \left(0, \frac{\sigma_u^2 + \delta^2 \sigma_w^2}{\sigma_x^2 \rho_{xz}^2} \right)$$

□

I. 5 ASYMPTOTIC DISTRIBUTION OF $\hat{\beta}_{\text{RED}}$ (LEMMA 5)

Proof. The proof is straightforward: this is the usual bivariate, unbiased case, with an error term equal to $(\delta + \beta_1 \gamma)w_i + u_i + \beta_1 e_i$. Since w , u and ϵ_{RED} are uncorrelated, its variance is $(\delta + \beta_1 \gamma)^2 \sigma_w^2 + \sigma_u^2 + \beta_1^2 \sigma_e^2$. □